

Oasis

A study of variational quantum algorithms and their dynamical Lie algebras to analyze different ansatz and cost functions for barren plateau emergence and classical simulability

Studying and testing different variational quantum algorithms:

☒ VQLS

☐ QAOA

☐ QC-PINN (some implementation for a PDE? Not sure but its pretty interesting to think about variational layers in PINNs)

August - December

Task 1: Implement smallest scale

Background: I implemented a 3-qubit VQLS over the summer using PennyLane's guide to understand, but that was awhile so I think the plan is to rederive some of the math by hand to get a better understanding of the paper, then try and implement a 2 qubit version. A main point of understand as well is how the measurements and shots work, and a way to calculate/understand variatiance and deviation

Lit review: Read over VQLS paper again, take notes-ish, re-derive cost function math

VQLS Paper notes:

for a quantum linear systems problem (QLSP),

$$A|0\rangle = |b\rangle$$

Where A can be decomposed into unitary matrices,

$$A = \sum_{l=1}^L c_l A_l, \text{ where } A_l \in SU(2)$$

and $|b\rangle$ can be composed by some unitary

$$U|0\rangle = |b\rangle$$

This is similar to the assumption with the Variational Quantum Eigensolver where the Hamiltonian (observable measured to get the energy eigenstates) is given as a linear combination of Pauli operators,

$$H = \sum c_l \sigma_l$$

Where L is a polynomial function of the # of qubits (system size).

To solve this problem, VQLS uses an ansatz $V(\vec{\alpha})$ s/t

$$V(\vec{\alpha})|0\rangle = |x(\vec{\alpha})\rangle = |x\rangle$$

$\vec{\alpha}$ is a parameter vector of the ansatz which, as an input to a quantum computer, prepares a potential solution state and runs an efficient circuit that estimates some cost function, $C(\vec{\alpha})$. In simple terms, $C(\vec{\alpha})$ measures how much the component $A|x\rangle$ is orthogonal to $|b\rangle$.

Cost functions

A simple cost function just measures the overlap of the projector

$$|\psi\rangle\langle\psi|, |\psi\rangle = A|x\rangle$$

with $|b\rangle$. AKA:

$$\hat{C}_G = \text{Tr}(|\psi\rangle\langle\psi|(\mathbb{I} - |b\rangle\langle b|)) = \langle x|H_g|x\rangle$$

where

$$H_g = A^\dagger(\mathbb{I} - |b\rangle\langle b|)A$$

Aside for some nontrivial math: the use of the matrix trace here takes in two density matrices ($|\psi\rangle\langle\psi|$ and $|b\rangle\langle b|$) and outputs an overlap measure of the two states. This is known as the Hilbert-Schmidt inner product,

$$\langle\rho, \sigma\rangle_{\text{HS}} = \text{Tr}(\rho\sigma)$$

$$\rho = |\psi\rangle\langle\psi|, \sigma = |\varphi\rangle\langle\varphi|$$

This cost function is small if $|\psi\rangle$ is nearly proportional to $|b\rangle$ or if the norm of $|\psi\rangle$ is small. This leads to the cost function being divided by the norm of $|\psi\rangle$ to get

$$C_G = \hat{C}_G / \langle\psi|\psi\rangle = 1 - |\langle b|\psi\rangle|^2$$

Where bold psi is a normalized state. Global cost functions like this however tend to exhibit barren plateaus when training, so to improve, you can use local cost functions instead, using H_L rather than H_G , where the effective local hamiltonian is

$$H_L = A^\dagger U \left(\mathbb{I} - \frac{1}{n} \sum_{j=1}^n |0_j\rangle\langle 0_j| \otimes \mathbb{I}_j \right) U^\dagger A$$

The global cost function can be broken down like the following:

$$\begin{aligned} C_G &= 1 - |\langle b|\psi\rangle|^2 \\ &= 1 - \frac{|\langle 0|U^\dagger A V|0\rangle|^2}{\langle 0|V^\dagger A^\dagger A V|0\rangle} \\ &= 1 - \frac{\langle 0|U^\dagger A V|0\rangle \langle 0|V^\dagger A U|0\rangle}{\langle 0|V^\dagger A^\dagger A V|0\rangle} \end{aligned}$$

where P replaces the bolded $|0\rangle\langle 0|$,

$$P = \frac{1}{2} - \frac{1}{2n} \sum_j^{n-1} Z_j$$

This can be derived from the local hamiltonian, H_L , specifically the inner portion,

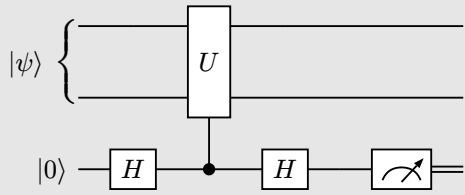
$$\frac{1}{n} \sum_j^n |0_j\rangle \langle 0_j| \otimes \mathbb{I}_{\bar{j}}$$

where j is the qubit and $\bar{j}$ is all other qubits.

$$\begin{aligned} |0\rangle \langle 0| &= \frac{\mathbb{I} - Z}{2} \Rightarrow \frac{1}{n} \sum_j^n \frac{\mathbb{I}_j - Z_j}{2} \otimes \mathbb{I}_{\bar{j}} \\ &= \frac{1}{n} \left(\sum_j^n \frac{1}{2} \mathbb{I}_j \otimes \mathbb{I}_{\bar{j}} - \frac{1}{2} Z_j \otimes \mathbb{I}_{\bar{j}} \right) \\ &= \frac{1}{n} \left(\sum_j^n \frac{1}{2} \mathbb{I} - \sum_j^n \frac{1}{2} Z_j \right) \\ &= \frac{1}{n} \left(\frac{n}{2} \mathbb{I} - \frac{1}{2} \sum_j^n Z_j \right) \\ &= \frac{1}{2} - \frac{1}{2n} \sum_j^n Z_j \quad (\text{where } \mathbb{I} \text{ is assumed}) \end{aligned}$$

Any cost function can be evaluated using a Hadamard-Test Circuit, where the real part of an expectation value of any hamiltonian can be extracted with some classical postprocessing.

For some observable, U , the **Hadamard-Test**, is a way to extract the expectation value of some observable by directly measuring a circuit by use of controlled gates and an ancilla (extra) qubit



Measuring the bottom qubit of this circuit you get

$$P(0) = \frac{1}{4} (2 + \langle \psi | U + U^\dagger | \psi \rangle)$$

and

$$P(1) = \frac{1}{4} (2 - \langle \psi | U + U^\dagger | \psi \rangle)$$

where the expectation value is

$$\begin{aligned} \mathbb{E} &= P(0) - P(1) = \frac{1}{2} \langle \psi | U + U^\dagger | \psi \rangle \\ &= \frac{1}{2} \text{Re}(\langle \psi | U | \psi \rangle) \end{aligned}$$

The imaginary part of the expectation value can also be extracted by applying a phase gate of $e^{-i\frac{\pi}{2}}$ to $|\psi\rangle$ before the control.

Ansatz

In the VQLS algorithm, $|x\rangle$ is prepared by acting on the zero state with a trainable gate sequence, $V(\vec{\alpha})$, with $\vec{\alpha}$ being a vector of continuous parameters. Typically the ansatz is dependent on gate overhead in the hardware that the algorithm is implemented on, known as a Hardware-Efficient Ansatz.

In addition to the HEA, you can also use the Quantum Alternating Operator Ansatz (QAOA), which takes the $H^{\otimes n}|0\rangle$ state (equal superposition over all qubits) and evolve it by some two hamiltonians for a number of layers.

To evolve a state by a hamiltonian means to use the unitary gates generated by the hamiltonians, H_D, H_M

$$e^{-i\alpha_i H_D}, e^{-i\alpha_j H_M}$$

where α_i is a continuous parameter and i is the round in total rounds, p , where the ansatz is

$$V(\vec{\alpha}) = \prod_l^{2p} e^{-i\alpha_l H_D} e^{-i\alpha_{l+1} H_M}$$

where the generated unitaries alternate hamiltonians to form an ansatz.

Training algorithm

In the paper they mention that there are multiple different classical optimizers you could use to train $V(\vec{\alpha})$. One of these algorithms (which I later implement for the 10-qubit VQLS) uses a method that, at each iteration, chooses a random direction, $\vec{w}$ in the parameter space and performs a line search, solving $\min_{s \in \mathbb{R}} C(\vec{\alpha} + s\vec{w})$. They also talk about gradient-based methods and derive the gradients of the cost functions (local and global) to show that the same circuits that calculate cost functions can be used to compute gradients. For the QAOA ansatz aswell, specific training is used.

Resilience to noise

In the paper they mention how some VQAs have been shown to be resilient to noise, in the way that the optimal parameters for the algorithms are unaffected by different noise models, known as Optimal Parameter Resilience (OPR). In the appendix they show that VQLS exhibits OPR with the normalized cost functions. Both being resilient to global depolarizing noise and C_L being resilient to measurement noise (good because C_L is the cost function actually used for implementations at scale). In the paper they talk about an error mitigation procedure called Probabilistic Error Cancellation (PEC) in which they represent an ideal gate through a basis of noisy gates that a quantum computer can implement. Basically they show that for certain noise models, VQLS can still obtain the optimal solution.

Hadamard Overlap-Test